

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 314 — Past Questions & Materials
by Renance

Course: MTH 314
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 5
Levels: 300

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Paper 1 of 5 — MTH 314 Metric Space Topology 2020 2021 First Semester

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

300 COLLEGE OF SCIENCE

— DEPARTMENT OF MATHEMATICS/COMPUTER SCIENCE

* FIRST SEMESTER EXAMINATION FOR 2020/2021 ACADEMIC SESSION

MTH 314: METRIC SPACE TOPOLOGY TIME ALLOWED: 2 HRS:30 MINS

| INSTRUCTION: ANSWER FOUR (4) QUESTIONS ONLY.

QUESTION ONE

a) Define the term Metric Space. (4 = marks)

b) Let $X = \mathbb{R}$ and let $d: \mathbb{R} \times \mathbb{R} \rightarrow [0, \infty)$ defined by $d(x_1, x_2) = |x_1 - x_2|$. Show that d is a metric on $\mathbb{R}$. (8 marks)

c) Let X be any set and $d: X \times X \rightarrow [0, \infty)$ be defined by $d(x, y) = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases}$. Show that d is a metric on X . (5 marks)

QUESTION TWO

a) Define (i) Open Ball (ii) Open subset (iii) Closed subset (4 = marks)

b) State and prove a theorem which describes the most important and basic properties of the collection of open subsets of a metric space (8 marks)

c) State a theorem which describes the most important and basic properties of the collection of closed subsets of a metric space (5 marks)

QUESTION THREE

a) Let (X, d) and (Y, d') be metric spaces, and let $f: X \rightarrow Y$ be a set map. When is f said to be continuous? (4 marks)

b) Show that a map $f: (X, d) \rightarrow (Y, d')$ between two metric spaces is continuous if and only if for any open set $U \subset Y$, $f^{-1}(U)$ is open in X . (6 marks)

c) Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be continuous maps of metric spaces. Show that the composition map $(g \circ f): X \rightarrow Z$ is a continuous map. (7 marks)

QUESTION FOUR

a) Let (X, d) be a metric space and let $A \subset X$. Define the terms (i) Interior of A (ii) Closure of A . (iii) Boundary of A . (4 = marks)

(b) Let (X, d) be a metric space. Show that if the sequence $\{x_n\}_{n \in \mathbb{N}}$ in X has a limit, then it is unique. (6 marks)

(c) Let (X, d) be a metric space and let $V \subset X$ be a subset. Show that the following statements are equivalent:

(i) V is a closed subset of X , (ii) the limit of every convergent sequence $\{x_n\}_{n \in \mathbb{N}}$ in V is an element of V . (7

marks)

QUESTION SIX

1

a) What do you understand by connectedness of a set? Give 2 examples. (4- marks)

b) Let (X, d_x) be a metric space. Show that the following are equivalent. (i) (X, d_x) is connected. (ii) There

does not exist open subsets $U, V \subset X$ such that $U \cup V = X$ and $U \cap V = \emptyset$. (iii) There does not exist any continuous map $f: X \rightarrow \{0\} \cup \{1\}$ which is surjective. (9 marks)

c) State without proof Heine-Borel theorem. (4 marks)

QUESTION FIVE

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a) Define the Homeomorphism. (3 - marks)

b) Let $f: (X, d_x) \rightarrow (Y, d_y)$ be a map. Show that the following are equivalent. (i) f is a homeomorphism, (ii) f

is bijective, continuous and open, (iii) f is bijective, continuous and closed. (9 marks)

c) Discuss convergence of sequences in a set X with the discrete metric. (5 marks)

Paper 2 of 5 — MTH 314 321 Metric Space Topology 2018 2019 First Semester

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN COLLEGE.
DEPARTMENT OF MATHEMATICS & COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION 2018/2019 SESSION
COURSE CODE: MTH 314 \ MTH 321 ; COURSE TITLE: METRIC SPACE TOPOLOGY
INSTRUCTIONS: ANSWER ANY FOUR QUESTIONS: TIME 2½ HOURS

QUESTION ONE

- (a) Define a metric space
- (b) Let X be a nonempty set with $X = \mathbb{R}$ and the function d as $d(x, y) = \begin{cases} 0, & \text{if } x = y \\ 1, & \text{if } x \neq y \end{cases}$. Prove that d is a metric on X
- (c) Let Y be a nonempty set with $Y = \mathbb{R}^n$ and the function D defined by $D(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$, for all $x, y \in Y, n \in \mathbb{N}$. Prove that (D, Y) is a metric space.

QUESTION TWO

- (a) Given $f(x) = x$ and $g(x) = x^2$, for all $x \in [0, 1]$. Find $d(f, g) = |f - g|$ and show that $d(f, g)$ is a metric.
- (b) Let $f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ 1, & \text{if } x = 1 \end{cases}$ and $g(x) = \frac{2x}{3}$, for all $x \in [0, 1]$. Show that $\sup_{x \in [0, 1]} |f(x) - g(x)| = \frac{2}{3}$
- (c) Define a normed space

QUESTION THREE

- (a) Define the following: open ball, closed ball and unit ball.
- (b) Show that the interval $I = (0, 1)$ is open.
- (c) Prove that in any metric space (X, d) , the closed ball with center r is a closed set.

QUESTION FOUR

- (a) Define the following: limit point, isolated point and limit of a function.
- (b) Consider $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 2x + 1$. For $\mathbb{R}$, we have $d(x, y) = |x - y|$. Show that $\lim_{x \rightarrow 1} f(x) = 3$.
- (c) Consider $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1 + x_2$. Show that $\lim_{x \rightarrow x_0} f(x) = 0$.

QUESTION FIVE

- (a) Define the following: continuity at a point, continuous function.
- (b) Consider $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2$. Show that this function is continuous at all $x_0 \in \mathbb{R}$
- (c) Prove continuity everywhere for $f: \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = 2x^2 - 8x + 6$.

QUESTION SIX

- (a) Define a compact set
- (b) Show that the closed interval $I = [a, b]$ is compact.
- (c) Show that compact sets are bounded

Paper 3 of 5 — MTH 314 Real Analysis 11 2018 2019 First Semester

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

UTER SCIENCE

DEPARTMENT OF MATHEMATICS AND COMP

FIRST SEMESTER EXAMINATION 2018/2019

\ Course Code/Title: MTH 314 — Real Analysis I

Instructions: Attempt any 4 questions. Time Allowed: 2'2 Hours.

Question 1 (17 2 Marks)

(a) Define Pointwise convergence (3 marks)

(b) Foreach integern > 1 , $f_n(x) = x^n$ on $[0, 1]$

marks)

(c) Prove or disprove if the Pointwise limit of $\{f_n\}_{n \in \mathbb{N}}$ is continuous on $(0, 1]$. (6marks)

_prove that $\{f_n\}_{n \in \mathbb{N}}$ converges Pointwise on $[0,1]$ (8%

Question 2 (17 % Marks) ay ed

(a) Define the following terms (i) functions of bounded variation (ii) Lipchitz function (iii) | monotone increasing and decreasing functions (6marks)

(b) Let $f(x) = x$ and $g(x) = x + [x]$. Compute

(c) State without proof, integration by substitution me

J, fod o (x) . (8 % marks)

thod. (3marks)

Question 3 (17 % Marks)

(a) State the theorem Uniform Cauchy Criterion (4marks)

$(x) = \sum_{n=1}^{\infty} w_n x^n \in [0, 4]$, $n \geq 1$. Find the set D a subset of $[0, 4]$ th

of real valued function. (5marks)

(c) Show that $f_y(x) = \sum_{n=1}^{\infty} v_n x^n$ is uniform convergence on $(0,1)$. ©@

VY, marks)

Question 4 (17 % Marks)

a) When is a bounded fu

b) Prove that the function $f: [0, 1]$

and that $\int_0^1 f(x) dx = \frac{1}{2}$ (10 % marks)

c) State three (3) basics properties of Riemann Integral (3marks)

netion $f: [a,b] \rightarrow \mathbb{R}$ said to be Riemann Integrable? (4marks)

_s $\mathbb{R}$ defined by $f(x) = x^2$ for all $x \in [0,1]$ is Riemann Integrable

Question 5 (17 VY, Marks)

(a) What do you understand by radius of conver

$3n$

; 7)" find the interval of convergence of the series. (82 marks)

(b) Given the series $\sum_{n=1}^{\infty} x^n$

(c) Consider the series $\sum_{n=0}^{\infty} (-1)^n \cos^n x$. Show that the series converges uniformly on $[0,1]$ to a differentiable function and $f'(x) = \sin x$. (6marks)

sence of a power series? (3marks)

Question 6 (17 'A Marks)

(a) State the Weierstrass M-test (3marks)

$\sum_{n=1}^{\infty} x^n$

(b) Show that $f: [0,1] \rightarrow \mathbb{R}$ defined by the series $f(x) = \sum_{n=1}^{\infty} \frac{x^n}{n^2}$ converges uniformly (8 Y2 marks)

t^2

(c) Compute $\int_0^1 f(t) dt$ and show that $\lim_{n \rightarrow \infty} \int_0^1 f_n(t) dt = \int_0^1 f(t) dt$

2

$\sum_{n=1}^{\infty} \frac{1}{n^2}$

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$\int_0^1 f(t) dt$ (6marks)

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Paper 4 of 5 — MTH 314, 2020.

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS

ANALYSIS 'THEMATICS/COM!

VEE AN SEMESTER EXAMINATION FOR 2019/2020 ACADEMIC YEAR

> METRIC SPACE TOPOLOGY TIME GIVEN

INSTRUCTION: ANSWER FOUR (4) QUESTIONS OF YOUR CHOICE

DEFINITION, QUESTION ONE

© THE TERM METRIC SPACE. (45 MARKS)

b) THE

$X = \mathbb{R}$

SHOW AND LET $d: \mathbb{R} \times \mathbb{R} \rightarrow [0, \infty)$ DEFINED

BY $d(x, y) = |x - y|$. (8 MARKS) (C) 2019/2020 ACADEMIC YEAR

LET X BE AN

Y-SET AND $d: X \times X \rightarrow [0, \infty)$ HAVING THE

PROPERTY THAT d IS A METRIC ON X . (8 MARKS) (C) 2019/2020 ACADEMIC YEAR

: QUESTION TWO

b) LINE (i) OPEN BALL (ii) OPEN SUBSET (iii) CLOSED SUBSET (4 MARKS)

LET (X, d)

A METRIC SPACE AND LET $\mathcal{O}$ BE THE COLLECTION OF ALL OPEN SUBSETS OF X WITH RESPECT TO d . PROVE THE

THEOREM: (i) THE EMPTY SET, X , ARE ELEMENTS OF $\mathcal{O}$. (ii) IF $\{U_\alpha\}_{\alpha \in I}$ IS ANY FAMILY OF OPEN SUBSETS, THEN $\bigcup_{\alpha \in I} U_\alpha \in \mathcal{O}$. (iii) IF $\{U_1, \dots, U_n\}$ IS ANY FINITE FAMILY OF OPEN SUBSETS, THEN $\bigcap_{i=1}^n U_i \in \mathcal{O}$. (8 MARKS)

CONSIDER $\mathbb{R}$ WITH THE EUCLIDEAN METRIC. SHOW

THAT FOR EACH $u \in \mathbb{R}$, THE SUBSET $[u, \infty)$

IS A CLOSED SUBSET OF $\mathbb{R}$. (4 MARKS)

a) LET (X, d) BE A METRIC SPACE

AND LET $A \subseteq X$. DEFINE THE TERMS (i) INTERIOR OF A (ii) CLOSURE OF A ,

(iii) BOUNDARY OF A . (45 MARKS)

(b) LET (X, d) BE A METRIC SPACE. SHOW THAT IF THE SEQUENCE

$\{x_n\}$ CONVERGES TO x IN X , THEN

x IS THE ONLY LIMIT OF $\{x_n\}$. (6 MARKS)

(c) LET (X, d) BE A METRIC SPACE AND LET $V \subseteq X$ BE A SUBSET. SHOW THAT

(i) V IS A CLOSED SUBSET OF X , (ii) THE LIMIT OF EVERY CONVERGENT SEQUENCE

IN V IS IN V . (6 MARKS)

nce $\{x, y\}$ is an element of V . (7

marks)

QUESTION FOUR

a) Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f: X \rightarrow Y$ be a set map. When is f said to be continuous?

(43 marks)

b) Let (X, d_X) and (Y, d_Y) and $g: (Y, d_Y) \rightarrow (Z, d_Z)$ be continuous maps of metric spaces. Show that the composition map $(g \circ f): (X, d_X) \rightarrow (Z, d_Z)$ is a continuous map of metric spaces. (7 marks)

c) Prove that the map $u: \mathbb{R} \rightarrow \mathbb{R}$ defined by $u(a, b) = a - b$ is a continuous function with respect to Euclidean metrics on $\mathbb{R}^2$ and $\mathbb{R}$. (6 marks)

QUESTION FIVE

1

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Define the Homeomorphism. (35 marks)

Show that the following are equivalent. (i) f is a homeomorphism,

Let $f: (X, d_X) \rightarrow (Y, d_Y)$ be a map. Show that

(i) f is bijective and

continuous and open, (iii) f is bijective, continuous

and maps sequences in a set X with the discrete metric. (5 marks)

QUESTION SIX

Define. (4 marks)

Define connectedness of a set? Give 2 examples ;

Show that the following are equivalent. (i) (X, d_X) is a uniform space (ii) =

subsets $U, V \subset X$ such that $U \cap V = \emptyset$ and $U \cup V = X$. (iii) There does not exist any

function $f: X \rightarrow \mathbb{R}$ which is surjective. (9 marks)

Define $\{0\}$ Borel theorem. (4 marks)

Paper 5 of 5 — MTH 314, 2019

Level 300 • First Semester • PastQuestion

QUESTION ONE

(a) Define a metric s ;

that d is a metric on X

(c) Let Y be a nonempty

subset of $\mathbb{R}^n$

Define $T: Y \rightarrow \mathbb{R}^n$ and the function D by

$D(x, y) = \sum_{i=1}^n |x_i - y_i|$, for all $x, y \in Y$

Prove that D is a metric on Y .

QUESTION TWO

(a) Given $f(x) = x$ and $\epsilon > 0$

find $\delta > 0$ such that $|f(x) - f(y)| < \epsilon$ whenever $|x - y| < \delta$

(b) Let $f(x) = \sin x$

on $[0, 1]$. Show that f is continuous on $[0, 1]$.

(c) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$f(x) = \begin{cases} x & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$

Prove that f is continuous at $x = 0$.

QUESTION THREE

(a) Define the following: open ball, closed ball and unit ball.

(b) Show that the interval $J = (0, 1)$ is open.

(c) Prove that in any metric space (X, d) , the closed ball with center r is a closed set.

□

QUESTION FOUR

(a) Define the following: limit point, isolated point

(b) Consider $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 2x + 1$

Prove that $\lim_{x \rightarrow 1} f(x) = 3$.

(c) Consider $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = x^2 + y^2$

Prove that f is continuous at $(0, 0)$.

1. For $d(x, y) = |x - y|$, show that

$d(x, y) \leq d(x, z) + d(z, y)$. Show that $\lim_{x \rightarrow 0} f(x) = 0$.

~ QUESTION FIVE

(a) Define the following: continuity at a point, continuous function.

(b) Consider $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2$. Show that this function is continuous at all $x \in \mathbb{R}$.

Prove that f is continuous at $x = 1$.

$f(x) = 2x^2 - 8x + 6$.

Prove continuity at $x = 1$.

□

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